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Table: 10.6 Differences between rod and cone cells

Rod cells	Cone cells
Rods are responsible for vision in dim light	The cones are responsible for colour vision and works best in the bright light.
The pigment present in the rods is rhodopsin, formed of a protein scotopsin and retinal (an aldehyde of vitamin A)	The pigment present in the cones is photopsin, formed of opsin protein and retinal.
There are about 120 millions rod cells	There may be 6-7 millions cone cells
Rods are predominant in the extra fovea region	Cones are concentrated in the fovea region

by using convex lens that converge the entering light rays on the retina.

Presbyopia: Due to aging, the lens loses elasticity and the power of accommodation. Convex lenses are used to correct this defect.

Astigmatism is due to the rough (irregular) curvature of cornea or lens. Cylindrical glasses are used to correct this error (Figure 10.17).

Cataract: Due to the changes in nature of protein, the lens becomes opaque. It can be corrected by surgical procedures.

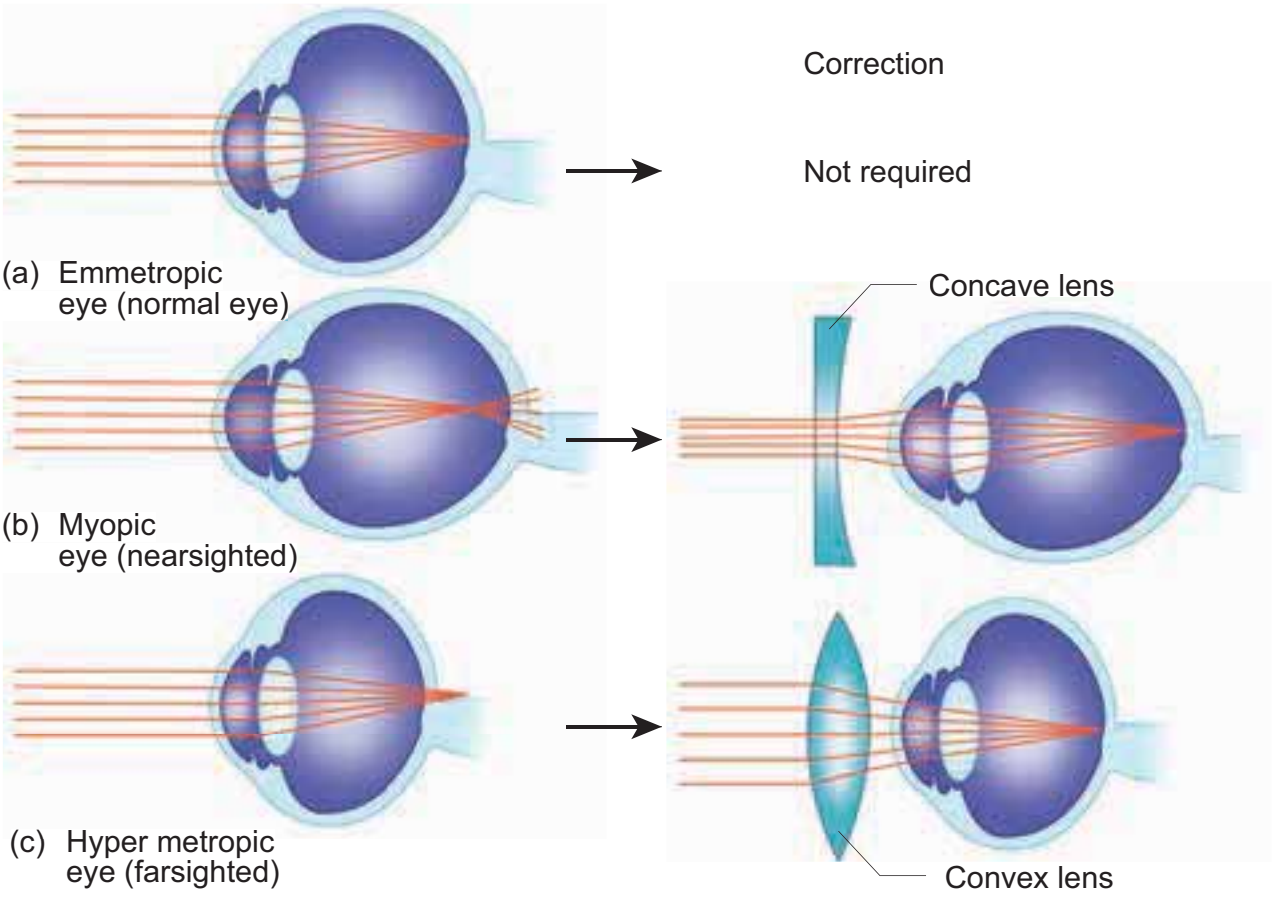


Figure 10.17 Refractive errors of the eye